Experiment 7

Integrating a Zoho Creator Application with Another Zoho Service using Connections and APIs

Low-Code Lab | Application: PATIENT APPOINTMENT SCHEDULING SYSTEM (Zoho Creator)

Name: Saranya V. Register No: 2116241501188 Department: AIML Rajalakshmi Engineering College

Abstract

This experiment adapts the Zoho integration procedure to the Patient Appointment Scheduling System. The proposed integration uses a Zoho OAuth connection and the Deluge invokeUrl task to send a notification to a Zoho Cliq channel when a new appointment is submitted. The screenshots in this report are illustrative mockups created for documentation; the connection and message delivery must be configured and tested in the live Zoho account.

1. Aim

To understand and design the integration of the Patient Appointment Scheduling System in Zoho Creator with Zoho Cliq using an OAuth connection and a REST API call.

2. Concepts

Connection

A saved, authorized link between Zoho Creator and another service. The user grants the required scopes and authorizes the connection.

API call

The Deluge invokeUrl task sends an HTTP request to the external service API. When a valid connection is supplied, Creator handles connection-based authorization.

Prebuilt integration

A ready-made integration action may be available in the workflow action picker. If the required service/action is not available, an API call can be used instead.

3. Integration Design

When a patient submits the Appointment Booking Form, the front desk should receive a notification in a Zoho Cliq channel named patientappointments. This notification is intended to supplement the appointment record and any confirmation process already configured.

Trigger: Appointment Booking Form → Created → Successful form submission.

4. Procedure

1. Open Zoho Cliq and create or select an organization/workspace that you are authorized to use.

2. Create a channel named patientappointments for appointment notifications.

3. In Zoho Creator, open Workflow and select Appointment Booking Form.

4. Create or edit a form workflow for Created → Successful form submission.

5. Add a Deluge script action. Create or select an OAuth connection named cliqconn for Zoho Cliq and authorize the required scopes supported by your account.

6. Use the script below as a template, then confirm the exact field link names, API endpoint domain, channel name, and connection scopes in your account.

7. Submit a test appointment and check whether the message appears in the Cliq channel. If the API returns an error, inspect the response and connection permissions.

5. Deluge Script Template

```
// Template: notify the front desk after a successful appointment submission
patientName = ifnull(input.Patient_Name, "Patient");
apptDate = input.Preferred_Appointment_Date.toString("dd-MMM-yyyy");
apptTime = input.Preferred_Appointment_Time.toString();

msg = "New appointment: " + patientName
    + " | Date: " + apptDate
    + " | Time: " + apptTime;

resp = invokeUrl
[
    url : "https://cliq.zoho.in/api/v2/channelsbyname/patientappointments/message"
    type : POST
    parameters : {"text": msg}.toString()
    connection : "cliqconn"
];
info resp;
```

Important: Patient_Name, Preferred_Appointment_Date and Preferred_Appointment_Time are example Deluge field link names based on the visible field labels. Use the actual field link names in your form. The API domain may vary by Zoho data center. Do not claim the integration is live until the connection is authorized and a test message is received.

6. Implementation Observations

The supplied application screenshots show enabled form workflows named Auto-Calculate Net Fee and Validate Treatment Consent, and an enabled Appointment Approval Workflow blueprint. They establish that the app already uses Creator workflow features. The supplied screenshots do not show an authorized Cliq connection or a successful live message.

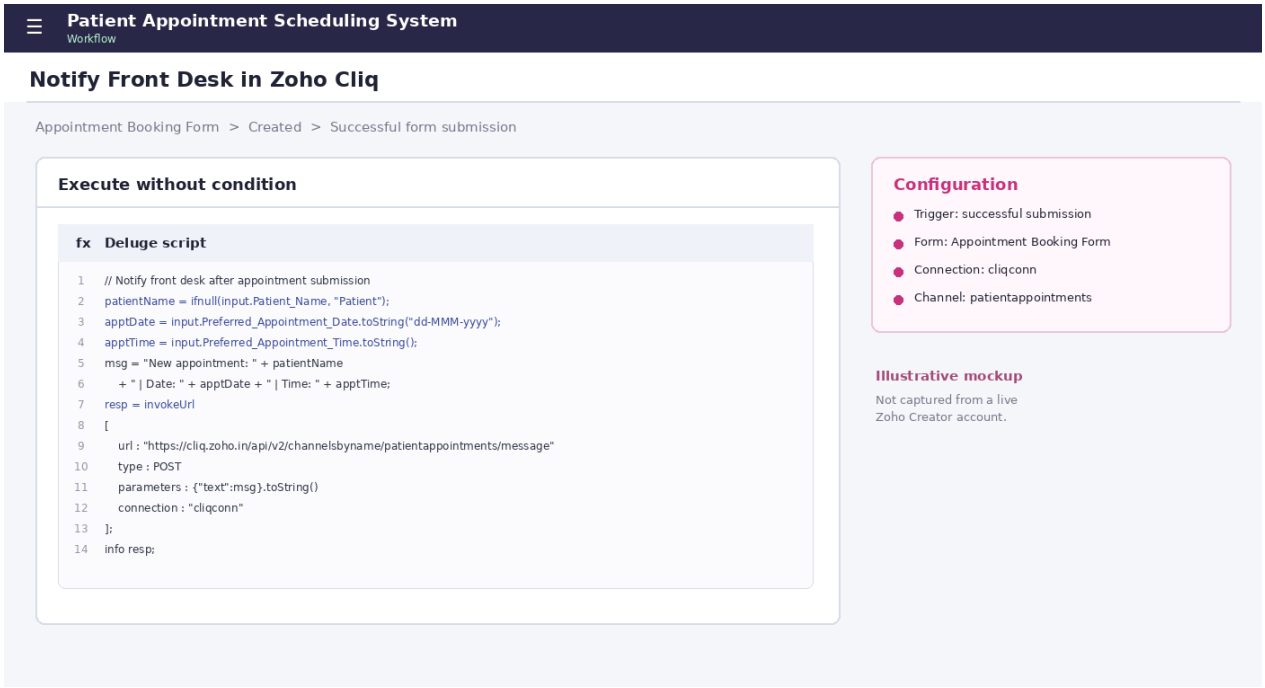


Figure 1: Illustrative workflow editor mockup for a successful appointment submission.

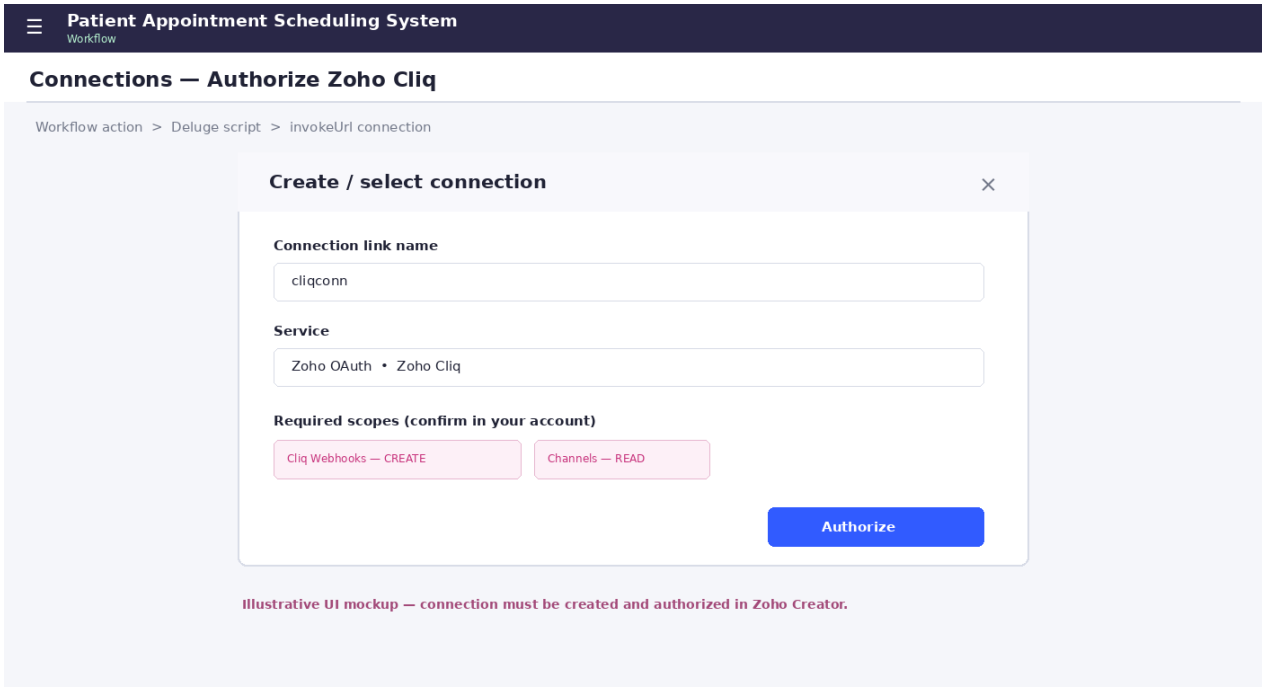


Figure 2: Illustrative connection setup mockup for a Zoho Cliq OAuth connection.

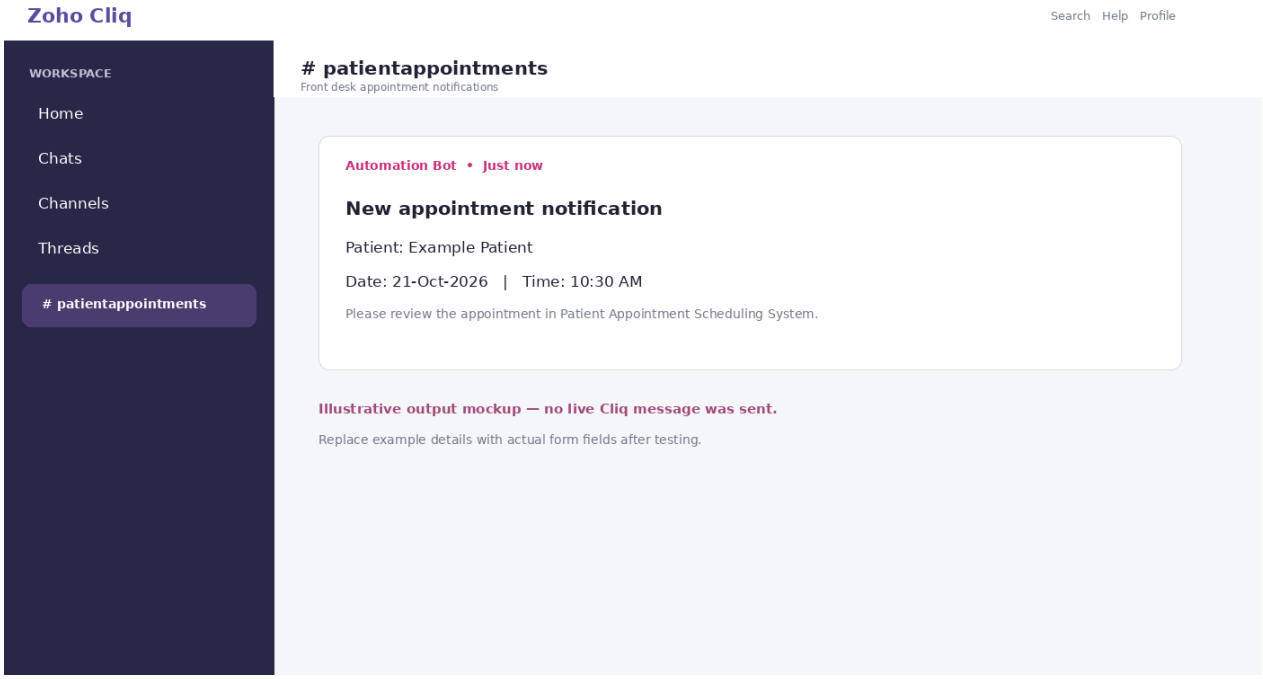


Figure 3: Illustrative sample Cliq notification output; not a live message.

7. Result

The Zoho Creator-to-Cliq integration design was documented for the Patient Appointment Scheduling System. The report describes the successful-submission trigger, OAuth connection setup, an invokeUrl script template, and a test procedure. The connection and end-to-end delivery remain to be completed and verified in the live account.

8. Learning Outcome

Learnt the role of OAuth connections, API calls through Deluge invokeUrl, workflow triggers, and service integration testing. Also learnt to distinguish a proposed integration design from a connection that has actually been authorized and tested.